

One-Pager Project Description (Group 5)

Problem Statement - We aim to predict a movie's IMDb rating based on its budget, director, actors, and other relevant information before the movie is released.

Dataset - We will use the IMDb datasets provided in the suggested sources. We have not finalized the features we will be focusing on but actors, writers, directors, runtime, and genre are potential features to focus on. An additional feature we are considering, which is potentially of high importance with a movie achieving a high rating, is budget. Due to the unavailability of the data related to budget and cost of movies in IMDb non-commercial database, we intend to use alternative sources such as TMDb and Kaggle to supplement the data from IMDb.

Machine Learning Model - Some of the features used in the model, such as movie genre and language, require one-hot encoding to convert string values into numerical values. We will use evaluation functions to generate values for attributes in each category (director, actors, etc) and a neural network or gradient descent approach to generate a model that can best represent the relation between these values and a final imdb rating.

Timeline -

4/27 1-pager topic description

Import data from the dataset into workable data structures (lists, vectors, dictionaries). Analyze the dataset and identify data points/features that require data cleaning or imputation (e.g., missing budget being replaced with the average budget for the movie genre).

During feature engineering, for complex features, such as actors/writers, goal is to identify a single coefficient representing the 'value' of a single contributor(actor) which consist of other sub features such as actor's popularity, previous movies or age.

Start exploratory data analysis and try to find correlations between individual features and rating.

5/2 Week

Continue exploratory data analysis and finalize what features to focus on based on the correlation analysis and other data exploration techniques. Begin developing a model or functions to evaluate individual categories

5/06 Mid-quarter project progress report

Based on the mid-quarter project report to justify our model again.

5/9 Week

Finish developing models to evaluate individual categories. Begin developing a model to combine individual categories and produce a final rating

Due 5/16 Week

Finish developing final model to predict the ratings. Begin testing model

Due 5/23 Week

Test and refine the final model ensuring accuracy

5/30 Week

Details: we will do the model test and refine the final model to ensure the accuracy rate. Based on the MSE, R^2 , Training_loss, Testing_loss, validation_loss vs epochs graphic, and cross validation technique to ensure our model has min loss and make sure our training dataset converges. Write a final project report, make a demo and push the source code on the github.

6/06 Project report, demo and source code.